

Rail clamp study - design brief

AI BRAKING / AB-SPEC / C0 / 2026-09-07

A rail-head clamp packaging study with opposed shoes and an actuator envelope. Rail dimensions, wedge/spring mechanism and certified holding force require an application-specific design.

TARGETS (not validated)

Illustrative rail head: 70 mm wide. Not a railway profile standard

Study holding force: 20 kN. Hypothesis

Assumed wet friction: 0.10. Sensitivity only; tests required

Normal force per shoe: 100 kN. $F / (2\mu)$, excluding mechanisms

DESIGN DECISIONS

Require the actual rail profile and wind-load analysis before developing a clamp mechanism.

Partner with a specialist for first installations.

Keep contamination and rail surface variation central to the requirements.

CANDIDATE MATERIALS

Frame: S355J2+N candidate only; loads may force a different section or material.

Rail shoes: supplier friction system; rail damage compatibility must be assessed.

Assembly and engineering gates

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ASSEMBLY INTENT

- 1. Obtain rail profile, rail restraint, wind loads and failure-state requirements.
- 2. Approve specialist mechanism and dimensional interface.
- 3. Finalize clamps, foundation, guarding and independent verification.

OPEN DESIGN RISKS

- 1. No functional wedge, latch or spring mechanism is modelled.
- 2. High required normal forces make this an application-specific structural programme.
- 3. No personnel-transport, storm safety or hazardous-area approval.

VALIDATION REQUIRED

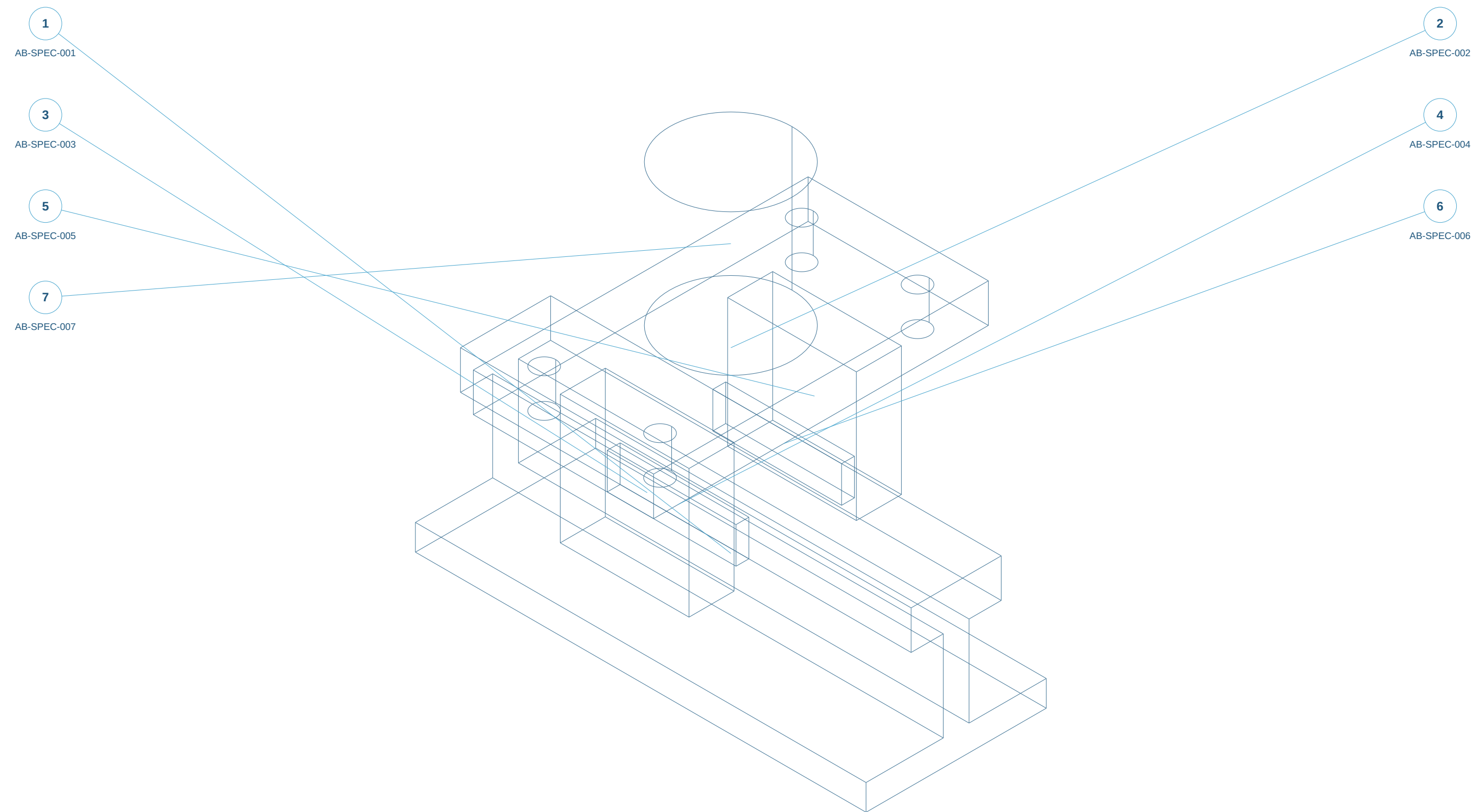
- 1. Wind load and machine/rail load path analysis.
- 2. Wet/oily rail holding tests, release clearance and fault testing.

ELECTRICAL / SENSOR REQUIREMENTS ONLY

- 1. Advisory position / wear sensing with replaceable brackets; no AI-controlled brake release.
- 2. Record supplier sensor range, supply, connector, EMC, ingress and temperature limits before selection.
- 3. Define independent machine safety interface and diagnostic fault response with the responsible controls engineer.

Assembly reference / numbered components

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Static isometric reference with item marks. Exploded geometry and part selection are available in the web workspace.

ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

Parts and interfaces

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AB-SPEC-001 | Rail reference envelope | interface-envelope

Customer rail specification required; Reference only; not supplied

AB-SPEC-002 | Clamp crosshead | make-concept

S355J2+N candidate; condition and design properties unapproved; Saw/laser blank; CNC critical faces and bores

AB-SPEC-003 | Jaw carrier -1 | make-concept

S355J2+N candidate; condition and design properties unapproved; Saw/laser blank; CNC critical faces and bores

AB-SPEC-004 | Rail shoe -1 | bought-envelope

Supplier-controlled; illustrative reserved envelope; Specialist rail-contact material

AB-SPEC-005 | Jaw carrier 1 | make-concept

S355J2+N candidate; condition and design properties unapproved; Saw/laser blank; CNC critical faces and bores

AB-SPEC-006 | Rail shoe 1 | bought-envelope

Supplier-controlled; illustrative reserved envelope; Specialist rail-contact material

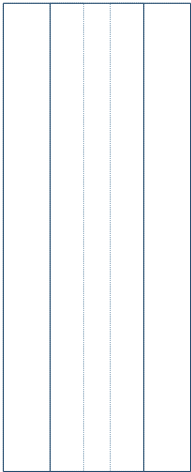
AB-SPEC-007 | Actuation reserved volume | interface-envelope

Supplier-controlled; illustrative reserved envelope; Specialist spring/wedge mechanism required

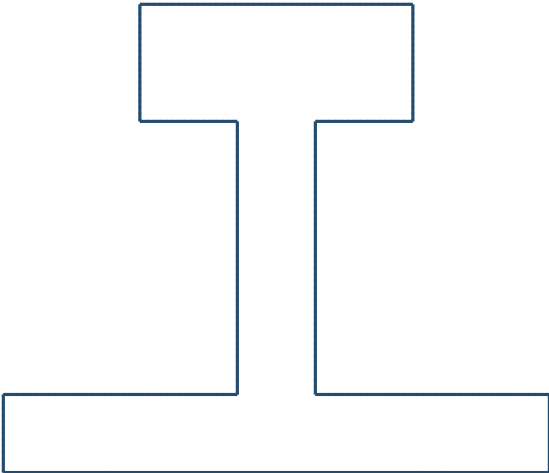
AB-SPEC-001 / Rail reference envelope

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TOP X-Y



FRONT X-Z



Customer rail specification required

Reference only; not supplied

Overall X/Y/Z bounds: 140.00 / 350.00 / 120.00 mm

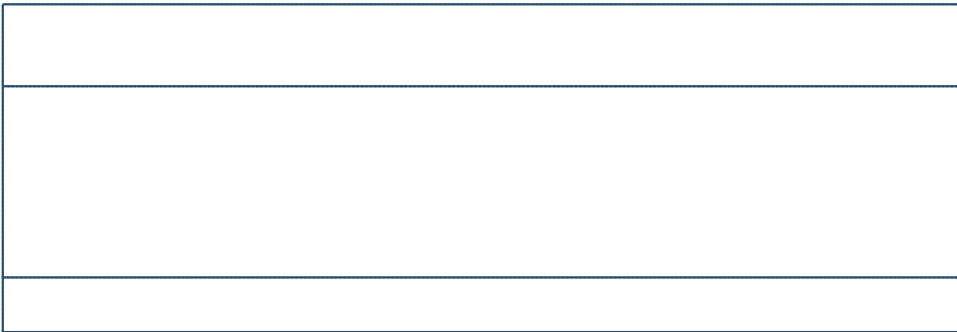
All dimensions nominal. Fits, GD&T and edge details awaiting engineer release.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

SIDE Y-Z



AB-SPEC-002 / Clamp crosshead

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TOP X-Y



FRONT X-Z



S355J2+N candidate; condition and design properties unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 260.00 / 140.00 / 30.00 mm

Original package interface only; not a structural sizing.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

SIDE Y-Z



AB-SPEC-003 / Jaw carrier -1

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TOP X-Y



FRONT X-Z



S355J2+N candidate; condition and design properties unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 35.00 / 100.00 / 100.00 mm

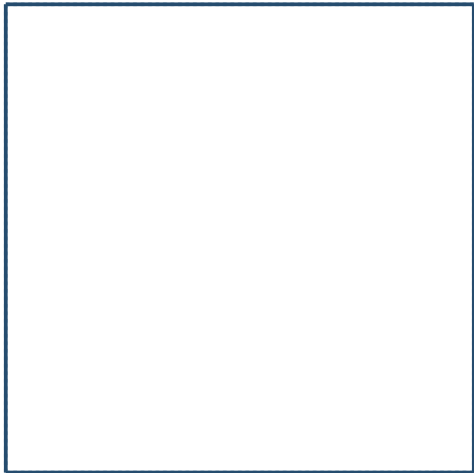
Carrier concept; mechanism, pins and adjustment omitted.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

SIDE Y-Z



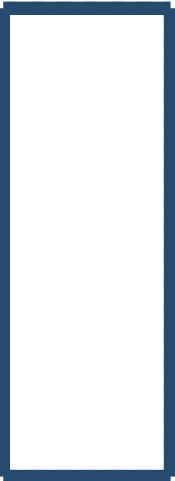
AB-SPEC-004 / Rail shoe -1

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TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Specialist rail-contact material

Overall X/Y/Z bounds: 10.00 / 100.00 / 28.00 mm

1 mm lateral rail clearance study.

No wedge/latch or contact force mechanism.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

SIDE Y-Z



AB-SPEC-005 / Jaw carrier 1

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TOP X-Y



FRONT X-Z



S355J2+N candidate; condition and design properties unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 35.00 / 100.00 / 100.00 mm

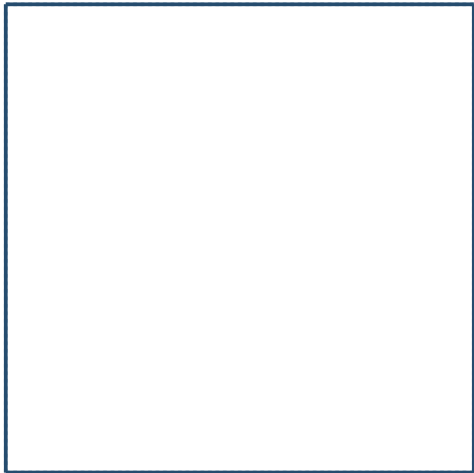
Carrier concept; mechanism, pins and adjustment omitted.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

SIDE Y-Z



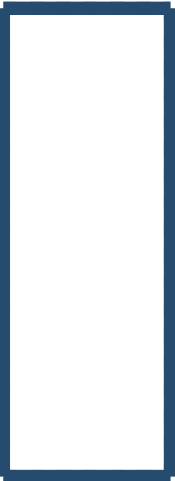
AB-SPEC-006 / Rail shoe 1

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TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Specialist rail-contact material

Overall X/Y/Z bounds: 10.00 / 100.00 / 28.00 mm

1 mm lateral rail clearance study.

No wedge/latch or contact force mechanism.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

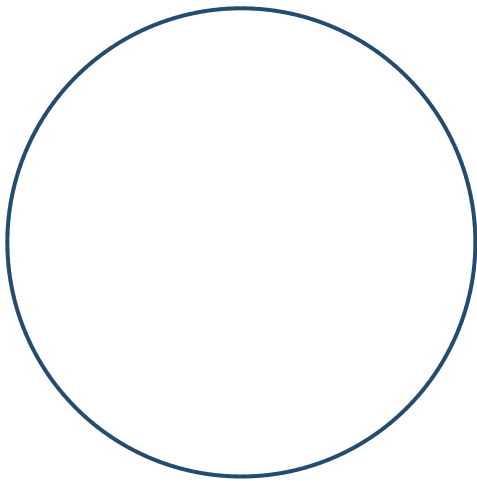
SIDE Y-Z



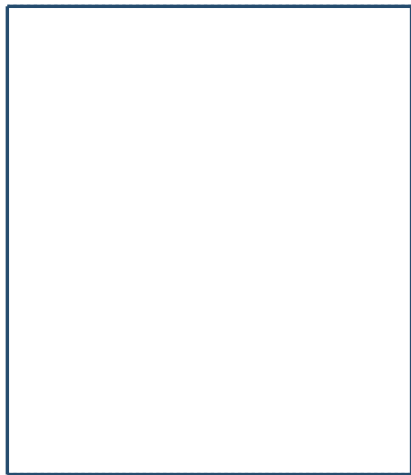
AB-SPEC-007 / Actuation reserved volume

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TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Specialist spring/wedge mechanism required

Overall X/Y/Z bounds: 95.00 / 95.00 / 110.00 mm

All dimensions nominal. Fits, GD&T and edge details awaiting engineer release.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

SIDE Y-Z

